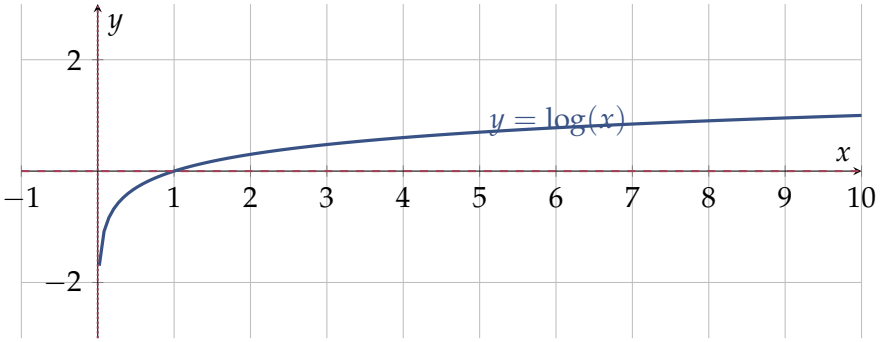
CHAPTER 3

Common logarithm function

Skills: converting between exponential and logarithmic form, solving, modelling

3.1 Reference notes

Key point. For every $b > 0$, $\log(b)$ is the unique real number x such that $10^x = b$. Hence $\log(10^x) = x$ and, for $b > 0$, $10^{\log(b)} = b$. The function log is defined on $]0, +\infty[$ and it is strictly increasing.



For $a > 0$, $b > 0$ and every natural number n:

$$\log(ab) = \log(a) + \log(b), \qquad \log\left(\frac{a}{b}\right) = \log(a) - \log(b), \qquad \log(a^n) = n\log(a).$$

Watch out. $\log(a + b)$ does not simplify. Before any logarithmic calculation, check that each argument is strictly positive.

Method. To solve $a^x = b$, with $a > 0$, $a \neq 1$ and $b > 0$, take the logarithm: $x\log(a) = \log(b)$, so $x = \dfrac{\log(b)}{\log(a)}$.

Example 1. Solving $1.08^x = 2$ amounts to writing $x = \dfrac{\log(2)}{\log(1.08)} \approx 9.01$. A value multiplied by 1.08 each year therefore doubles after a little more than nine years.

Since log is increasing, for $a, b > 0$, $a < b$ is equivalent to $\log(a) < \log(b)$. For $a > 1$, $a^x < b$ is equivalent to $x < \dfrac{\log b}{\log a}$; when $0 < a < 1$, the inequality sign reverses.

Session 15 — Definition of the common logarithm

Common logarithm function

17 Nov 2026 • B: 2 h – C: 1 h

Objectives.

- Link powers of 10 and logarithms.
- Calculate common values.
- Use the calculator.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 20 % of 80. 16
2. Give the domain of log. $]0, +\infty[$
3. Calculate $\log(10^4)$. 4
4. Solve $10^x = 0.01$. $x = -2$
5. Expand $\log(3b)$, $b > 0$. $\log 3 + \log b$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: link powers of 10 and logarithms and calculate common values.

Activity 1. Complete the pairs $10^x = b$ and $x = \log(b)$ for $b = 0.001; 0.1; 1; 10; 1\,000$. Then use the calculator for $b = 7$.

Solution. The logarithms are $-3, -1, 0, 1, 3$; $\log(7) \approx 0.8451$. Have students say aloud which piece of data triggers the choice of method.

Written record. For $b > 0$, $\log(b)$ is the unique solution of $10^x = b$. In particular $\log(1) = 0$, $\log(10) = 1$ and $\log(100) = 2$.

Example 2. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. The logarithms are $-3, -1, 0, 1, 3$; $\log(7) \approx 0.8451$.

Exercise 3.1 · essential.

1. Calculate $\log(10^2)$ and $\log(10^{-2})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $2^x = 22$ and give an approximate value to the nearest hundredth.

Solution. 2 and -2. The expression equals $2+2\log(a)-\log(b)$. Finally $x=\dfrac{\log(22)}{\log(2)}$.

Further practice: Calao, course pp. 64–69; exercises p. 70, no. 1–5. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Solve the following equations in $\mathbb{R}$. When a solution exists, give its exact value and then its value rounded to the nearest thousandth.

a. $4\times 10^{x}=9.2$

b. $7\times 10^{x}-1.5=3\times 10^{x}+0.9$

c. $2.5\times 10^{x}+12=6-0.5\times 10^{x}$

d. $10^{x+1}=50$ (you may write $10^{x+1}=10\times 10^{x}$)

Solution. **Target:** Isolate 10^{x}, then take the logarithm.

Method: gather the terms in 10^{x} on one side, isolate 10^{x}, check that the right-hand side is strictly positive, then use: for $b>0$, $10^{x}=b \iff x=\log(b)$.

a. $10^{x}=\dfrac{9.2}{4}=2.3>0$, so $x=\log(2.3)\approx 0.362$.

b. $7\times 10^{x}-3\times 10^{x}=0.9+1.5$, i.e. $4\times 10^{x}=2.4$, so $10^{x}=0.6>0$ and $x=\log(0.6)\approx -0.222$.

c. $2.5\times 10^{x}+0.5\times 10^{x}=6-12$, i.e. $3\times 10^{x}=-6$, so $10^{x}=-2$. But $10^{x}>0$ for every real x: the equation has no solution.

d. $10\times 10^{x}=50$, so $10^{x}=5>0$ and $x=\log(5)\approx 0.699$. (Another way: $x+1=\log(50)$, so $x=\log(50)-1=\log(5)$.)

Recommended calculator check: $10^{0.362}\approx 2.30$ and $10^{-0.222}\approx 0.60$.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 195 and the rate or common ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Notes

Session 16 — **Variations and graph**

Common logarithm function — 19 Nov 2026 • B: 1 h – C: 2 h

Objectives.

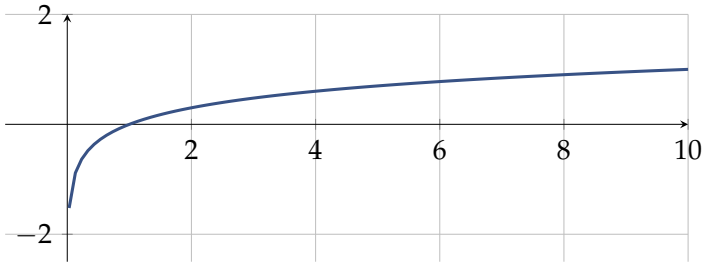
- Know the domain.
- Use the fact that the function is increasing.
- Read the curve.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for an increase of 25 %. $1 + \dfrac{25}{100} = 1.25$
2. Calculate $\log(10^0)$. 0
3. Solve $10^x = 0.01$. $x = -2$
4. Expand $\log(3b)$, $b > 0$. $\log 3 + \log b$
5. Give the domain of log. $]0, +\infty[$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: know the domain and use the fact that the function is increasing.

Activity 2. Plot $y = 10^x$ and the line $y = x$, then obtain the curve of the logarithm by symmetry. Deduce its domain, its direction of variation and its asymptote.



Solution. The logarithm is the inverse function of 10^x: domain $]0, +\infty[$, strictly increasing, asymptote $x = 0$. Have students say aloud which piece of data triggers the choice of method.

Written record. The function log is defined on $]0, +\infty[$, is strictly increasing, and its curve has $x = 0$ as a vertical asymptote.

Example 3. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. The logarithm is the inverse function of 10^x: domain $]0,+\infty[$, strictly increasing, asymptote $x = 0$.

Exercise 3.2 · essential.

1. Calculate $\log(10^3)$ and $\log(10^{-3})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $3^x = 23$ and give an approximate value to the nearest hundredth.

Solution. 3 and -3. The expression equals $2 + 2\log(a) - \log(b)$. Finally $x = \dfrac{\log(23)}{\log(3)}$.

Further practice: Calao, course pp. 64–69; exercises p. 70, no. 6–10. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Recall that the function log is strictly increasing on $]0;+\infty[$ and that $\log(1) = 0$.

1. Without a calculator, give the sign of each of the following numbers, with a justification:
   a. $\log(0.9)$;
   b. $\log\left(\dfrac{4}{3}\right)$;
   c. $\log(CM)$, where CM is the multiplier associated with an increase of $0.2\,\%$.
2. Without a calculator, compare $\log(0.35)$ and $\log(0.53)$.

Solution. **Target:** Sign of a logarithm and comparison.

1. For $a > 0$: if $a > 1$ then $\log(a) > \log(1) = 0$; if $0 < a < 1$ then $\log(a) < 0$ (since log is strictly increasing).
   a. $0 < 0.9 < 1$, so $\log(0.9) < 0$ (about -0.046).
   b. $\dfrac{4}{3} > 1$, so $\log\left(\dfrac{4}{3}\right) > 0$ (about 0.125).
   c. $CM = 1 + \dfrac{0.2}{100} = 1.002 > 1$, so $\log(1.002) > 0$ (about 0.000 87: very close to 0 but positive).
2. $0 < 0.35 < 0.53$ and log is strictly increasing on $]0;+\infty[$, so $\log(0.35) < \log(0.53)$. Both numbers are negative (arguments less than 1); the one closer to 0 is $\log(0.53)$.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 200 and the rate or common ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 17 — **Algebraic properties**

Common logarithm function — 24 Nov 2026 • B: 2 h – C: 1 h

Objectives.

- Expand or condense logarithms.
- Check the conditions.
- Avoid false properties.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 6 - x + 2$. $2x + 8$
2. Solve $10^x = 0.01$. $x = -2$
3. Expand $\log(3b)$, $b > 0$. $\log 3 + \log b$
4. Give the domain of log. $]0, +\infty[$
5. Calculate $\log(10^1)$. 1

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: expand or condense logarithms and check the conditions.

Activity 3. Starting from $10^2 \times 10^3 = 10^5$, conjecture $\log(ab) = \log a + \log b$. Test numerically with $a = 2, b = 5$, then explain why a, b must be positive.

Solution. $\log(2) + \log(5) = \log(10) = 1$. The logarithm is only defined for a strictly positive argument. Have students say aloud which piece of data triggers the choice of method.

Written record. For $a, b > 0$, $\log(ab) = \log a + \log b$ and $\log\left(\frac{a}{b}\right) = \log a - \log b$. Moreover $\log(a^n) = n \log a$.

Example 4. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. $\log(2) + \log(5) = \log(10) = 1$. The logarithm is only defined for a strictly positive argument.

Exercise 3.3 · essential.

1. Calculate $\log(10^4)$ and $\log(10^{-4})$.
2. Expand $\log\left(\frac{100a^2}{b}\right)$ for $a, b > 0$.

3. Solve $4^x = 24$ and give an approximate value to the nearest hundredth.

Solution. 4 and -4. The expression equals $2 + 2\log(a) - \log(b)$. Finally $x = \dfrac{\log(24)}{\log(4)}$.

Further practice: Calao, course pp. 66–69; exercises p. 72, no. 20–23. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Write each of the following numbers in the form $\log(A)$, where A is a strictly positive real number to be given as an integer or a fraction. When possible, give the exact value of the number.

a. $\log(6) + \log(5) - \log(3)$

b. $2\log(4) - \log(8)$

c. $\log(5) + \log(11)$

d. $2\log(3) - 3\log(2)$

e. $\log(40) - \log(8) + 2\log(2)$

Solution. **Target:** Condense an expression into one logarithm.

We use, for $a, b > 0$ and any natural number n: $\log(a) + \log(b) = \log(ab)$, $\log(a) - \log(b) = \log\left(\dfrac{a}{b}\right)$ and $n\log(a) = \log(a^n)$. All the arguments are strictly positive.

a. $\log(6) + \log(5) - \log(3) = \log\left(\dfrac{6 \times 5}{3}\right) = \log(10) = 1.$

b. $2\log(4) - \log(8) = \log(4^2) - \log(8) = \log\left(\dfrac{16}{8}\right) = \log(2).$

c. $\log(5) + \log(11) = \log(55).$

d. $2\log(3) - 3\log(2) = \log(3^2) - \log(2^3) = \log\left(\dfrac{9}{8}\right).$

e. $\log(40) - \log(8) + 2\log(2) = \log\left(\dfrac{40 \times 2^2}{8}\right) = \log(20).$

Error to point out: $\log(5) + \log(11)$ is not $\log(16)$; a sum of logarithms corresponds to a product of the arguments.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 205 and the rate or common ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 18 — Solving $10^x = b$

Common logarithm function

26 Nov 2026 • B: 1 h – C: 2 h

Objectives.

- Solve $10^x = b$.
- Bracket a solution.
- Interpret an order of magnitude.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 5 % of 140. 7
2. Expand $\log(3b)$, $b > 0$. $\log 3 + \log b$
3. Give the domain of log. $]0, +\infty[$
4. Calculate $\log(10^2)$. 2
5. Solve $10^x = 0.01$. $x = -2$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: solve $10^x = b$ and bracket a solution.

Activity 4. A device measures intensities $b = 0.04; 3.2; 850$. For each value, solve $10^x = b$, give x to the nearest hundredth and check by calculating 10^x.

Solution. $x = \log(b)$: approximately -1.40, 0.51 and 2.93. Have students say aloud which piece of data triggers the choice of method.

Written record. The equation $10^x = b$ has a solution only if $b > 0$; this solution is $x = \log(b)$.

Example 5. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. $x = \log(b)$: approximately -1.40, 0.51 and 2.93.

Exercise 3.4 · essential.

1. Calculate $\log(10^5)$ and $\log(10^{-5})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $5^x = 25$ and give an approximate value to the nearest hundredth.

Solution. 5 and -5. The expression equals $2+2\log(a)-\log(b)$. Finally $x=\dfrac{\log(25)}{\log(5)}$.

Further practice: Calao, p. 73, no. 30–32. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A delivery company buys a van for 38 000 €. Its value is estimated to fall by 15 % each year. The company will sell the van as soon as its value becomes strictly less than 10 000 €. Let V be the value of the van (in euros) and N the number of years since the purchase.

```
V ← 38 000
N ← 0
While V .....
    V ← .....
    N ← N + 1
End While
```

1. Complete the algorithm so that, at the end of its execution, the variable N contains the number of years after which the van will be sold.
2. Using the calculator (table of values), run the algorithm and find the value of N obtained. Give the value of the van when it is sold, rounded to the nearest euro.
3. Write this program in Python, run it and compare with the result of question 2.

Solution. **Target:** Threshold algorithm for a change over time.

1. A 15 % decrease corresponds to the multiplier $1-0.15=0.85$. We repeat as long as the value has not yet fallen below the threshold: "While $V \geqslant 10\,000$" and "$V \leftarrow 0.85 \times V$".
2. Successive values rounded to the nearest euro: $N=6$: 14 332 €; $N=7$: 12 182 €; $N=8$: 10 355 €; $N=9$: 8 801 €. Since $38\,000 \times 0.85^8 \approx 10\,355 \geqslant 10\,000$ and $38\,000 \times 0.85^9 \approx 8\,801 < 10\,000$, the loop stops with $N=9$: the van will be sold after 9 years, for a value of about 8 801 €.
3. A possible program:

```
V = 38000
N = 0
while V >= 10000:
    V = 0.85 * V
    N = N + 1
print(N, V)
```

The program displays 9 and about 8801.44, which confirms question 2. Stress the decimal point in Python and the indentation of the loop block.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 210 and the rate or common ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 19 — Solving $a^x = b$

Common logarithm function

1 Dec 2026 • B: 2 h – C: 1 h

Objectives.

- Solve $a^x = b$.
- Use a quotient of logarithms.
- Round sensibly.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for an increase of 10 %. $1 + \dfrac{10}{100} = 1.10$
2. Give the domain of log. $]0, +\infty[$
3. Calculate $\log(10^3)$. 3
4. Solve $10^x = 0.01$. $x = -2$
5. Expand $\log(3b)$, $b > 0$. $\log 3 + \log b$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: solve $a^x = b$ and use a quotient of logarithms.

Activity 5. The number of subscribers is multiplied by 1.15 each month. Determine after how many months it will have tripled. Set up the equation, solve it using logarithms and interpret the rounding.

Solution. $1.15^n \geq 3$; $\dfrac{\log 3}{\log 1.15} \approx 7.86$. So eight whole months are needed. Have students say aloud which piece of data triggers the choice of method.

Written record. For $a > 0$, $a \neq 1$ and $b > 0$, $a^x = b$ is equivalent to $x = \dfrac{\log(b)}{\log(a)}$.

Example 6. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. $1.15^n \geq 3$; $\dfrac{\log 3}{\log 1.15} \approx 7.86$. So eight whole months are needed.

Exercise 3.5 · essential.

1. Calculate $\log(10^6)$ and $\log(10^{-6})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $6^x = 26$ and give an approximate value to the nearest hundredth.

Solution. 6 and -6. The expression equals $2 + 2\log(a) - \log(b)$. Finally $x = \dfrac{\log(26)}{\log(6)}$.

Further practice: Calao, p. 74, no. 33–39. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. An online shop A received 2 500 orders in January 2026. This number is assumed to increase by 7 % per month. Let a_n be the number of orders n months after January 2026.

1. a. Express a_n in terms of n.
   b. After how many months will the monthly number of orders have doubled?
   c. After how many months will it have quadrupled? Is this result twice the previous one? Explain.
2. A competing shop B received 4 000 orders in January 2026, with an increase of 3 % per month. From how many months on will shop A have more monthly orders than shop B? In which month will this happen?

Solution. **Target:** Doubling time and overtaking.

1. a. An increase of 7 % corresponds to the multiplier 1.07: $a_n = 2\,500 \times 1.07^n$.
   b. $a_n \geqslant 5\,000 \iff 1.07^n \geqslant 2 \iff n\log(1.07) \geqslant \log(2) \iff n \geqslant \dfrac{\log(2)}{\log(1.07)}$, because $\log(1.07) > 0$ (the inequality keeps its direction). Now $\dfrac{\log 2}{\log 1.07} \approx 10.24$: 11 months are needed. Check: $1.07^{10} \approx 1.967 < 2$ and $1.07^{11} \approx 2.105 \geqslant 2$.
   c. $1.07^n \geqslant 4 \iff n \geqslant \dfrac{\log(4)}{\log(1.07)} \approx 20.49$: 21 months are needed ($1.07^{20} \approx 3.870$ and $1.07^{21} \approx 4.141$). Since $\log(4) = 2\log(2)$, the exact threshold is indeed doubled ($20.49 = 2 \times 10.24$), but rounding up to the next integer happens only once: 21, not $2 \times 11 = 22$.
2. We look for n such that $2\,500 \times 1.07^n > 4\,000 \times 1.03^n$. Dividing by $2\,500 \times 1.03^n > 0$: $\left(\dfrac{1.07}{1.03}\right)^n > 1.6$. Since $\dfrac{1.07}{1.03} > 1$, its logarithm is positive and $n > \dfrac{\log(1.6)}{\log(1.07/1.03)} \approx 12.34$. Hence $n = 13$: check $a_{12} \approx 5\,630 < 5\,703$ and $a_{13} \approx 6\,025 > 5\,874$. Shop A will overtake shop B in February 2027 (13 months after January 2026).

Homework. Redo practice exercise 2 with the following data: replace the initial value by 215 and the rate or common ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 20 — **Equations $x^a = b$ and simple inequalities**

Common logarithm function — 3 Dec 2026 • B: 1 h – C: 2 h

Objectives.

- Solve equations where the variable is the base.
- Deal with an inequality.
- Take the domain into account.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 9 - x + 2$. $2x + 11$
2. Calculate $\log(10^4)$. 4
3. Solve $10^x = 0.01$. $x = -2$
4. Expand $\log(3b)$, $b > 0$. $\log 3 + \log b$
5. Give the domain of log. $]0, +\infty[$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: solve equations where the variable is the base and deal with an inequality.

Activity 6. Solve on $[0, +\infty[$: $x^3 = 40$, then $x^4 = 16$. Then solve $1.08^x < 1.5$ and explain the role of the function being increasing.

Solution. $x = \sqrt[3]{40} \approx 3.42$; $x = 2$ on the requested interval; $x < \dfrac{\log 1.5}{\log 1.08} \approx 5.27$. Have students say aloud which piece of data triggers the choice of method.

Written record. To solve $x^n = b$, take the domain and the parity of n into account. For an exponential inequality, the direction depends on whether the function is increasing or decreasing.

Example 7. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. $x = \sqrt[3]{40} \approx 3.42$; $x = 2$ on the requested interval; $x < \dfrac{\log 1.5}{\log 1.08} \approx 5.27$.

Exercise 3.6 · essential.

1. Calculate $\log(10^2)$ and $\log(10^{-2})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.

3. Solve $2^x = 22$ and give an approximate value to the nearest hundredth.

Solution. 2 and -2. The expression equals $2 + 2\log(a) - \log(b)$. Finally $x = \dfrac{\log(22)}{\log(2)}$.

Further practice: Calao, p. 75, no. 40–42. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. The retail price of a solar panel model went from 420 € in March 2021 to 260 € in September 2025, a period of 4 years and 6 months. The price is assumed to have changed at a constant annual rate t.

1. Justify that $(1+t)^{4.5} = \dfrac{260}{420}$.
2. Deduce t, rounded to 0.1 %. Interpret the sign of the result.
3. A second model costs 395 € today. If its price changes at the same annual rate, what will its price be in 2 years and 3 months? Round to the nearest euro.
4. The manufacturer of this second model would rather reach a price of 300 € in 2 years and 3 months. What constant annual rate should it apply? Round to 0.1 %.

Solution. **Target:** Finding a rate: equation $x^n = b$.

1. Over 4.5 years, the price is multiplied by $(1+t)^{4.5}$, so $420(1+t)^{4.5} = 260$, i.e. $(1+t)^{4.5} = \dfrac{260}{420} \approx 0.619$.
2. $1 + t > 0$, so $1 + t = \left(\dfrac{260}{420}\right)^{1/4.5} \approx 0.8989$, hence $t \approx -0.101$, i.e. $t \approx -10.1\,\%$. The rate is negative: the price fell on average by about 10.1 % per year. Check: $420 \times 0.8989^{4.5} \approx 260$.
3. 2 years and 3 months $= 2.25$ years. Price: $395(1+t)^{2.25}$. Keeping the exact value of $1+t$ in memory: $395 \times \left(\dfrac{260}{420}\right)^{2.25/4.5} = 395 \times \left(\dfrac{260}{420}\right)^{0.5} \approx 395 \times 0.7868 \approx 310.78$. The price would be about 311 €.
4. We solve $395(1+t')^{2.25} = 300$, i.e. $(1+t')^{2.25} = \dfrac{300}{395} \approx 0.7595$, so $1 + t' = \left(\dfrac{300}{395}\right)^{1/2.25} \approx 0.8849$ and $t' \approx -11.5\,\%$. A decrease of about 11.5 % per year would be needed.

Remark: using a value of t rounded too early slightly changes the results (for example $395 \times 0.899^{2.25} \approx 310.85$); keep the value in memory.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 220 and the rate or common ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 21 — **Problems in context**

Common logarithm function — 8 Dec 2026 • B: 2 h – C: 1 h

Objectives.

- Model a situation.
- Choose an equation.
- Conclude in context.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 20 % of 60. 12
2. Solve $10^x = 0.01$. $x = -2$
3. Expand $\log(3b)$, $b > 0$. $\log 3 + \log b$
4. Give the domain of log. $]0, +\infty[$
5. Calculate $\log(10^3)$. 3

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: model a situation and choose an equation.

Activity 7. The sound level is $L = 10\log\left(\dfrac{I}{I_0}\right)$. Calculate L when $I = 10^6 I_0$. Then determine the ratio $\dfrac{I}{I_0}$ for $L = 85$ dB.

Solution. $L = 60$ dB. For 85 dB, $\log\left(\dfrac{I}{I_0}\right) = 8.5$, so $\dfrac{I}{I_0} = 10^{8.5} \approx 3.16 \times 10^8$. Have students say aloud which piece of data triggers the choice of method.

Written record. In a problem, define the variable, set up the equation, solve, check the domain and conclude with the unit. A numerical solution without interpretation is incomplete.

Example 8. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. $L = 60$ dB. For 85 dB, $\log\left(\dfrac{I}{I_0}\right) = 8.5$, so $\dfrac{I}{I_0} = 10^{8.5} \approx 3.16 \times 10^8$.

Exercise 3.7 · essential.

1. Calculate $\log(10^3)$ and $\log(10^{-3})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $3^x = 23$ and give an approximate value to the nearest hundredth.

Solution. 3 and -3. The expression equals $2 + 2\log(a) - \log(b)$. Finally $x = \dfrac{\log(23)}{\log(3)}$.

Further practice: Calao, p. 74, no. 38–39, then p. 75, no. 40–42. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. In a laboratory, a cell culture grows by successive divisions: at each generation, the number of cells doubles. When the number of cells goes from N_0 to N, the number of generations that have elapsed is

$$g = \frac{\log\left(\dfrac{N}{N_0}\right)}{\log(2)}.$$

For example, going from 1 to 8 cells corresponds to $g = \dfrac{\log(8)}{\log(2)} = 3$ generations.

1. A culture goes from 1 000 to 64 000 cells. How many generations have elapsed?
2. Another culture goes from 500 to 128 000 cells. Same question.
3. A culture of 1 000 cells develops for 9.5 generations. About how many cells does it contain?
4. Starting from a single cell, how many cells are obtained after 20 generations?
5. A laboratory starts from 1 000 cells and wants to obtain 1 000 000 000. How many generations are needed? Given that one generation lasts 30 minutes, how much time should be planned?

Solution. **Target:** Using a formula $\log(N)/\log(2)$.

1. $\dfrac{N}{N_0} = 64 = 2^6$, so $g = \dfrac{\log(2^6)}{\log(2)} = \dfrac{6\log(2)}{\log(2)} = 6$ generations.
2. $\dfrac{N}{N_0} = \dfrac{128\,000}{500} = 256 = 2^8$, so $g = 8$ generations.
3. $g = 9.5$ gives $\log\left(\dfrac{N}{1\,000}\right) = 9.5\log(2)$, so $\dfrac{N}{1\,000} = 10^{9.5\log(2)} = 2^{9.5} \approx 724.08$. Thus $N \approx 724\,000$ cells.
4. $N = 2^{20} = 1\,048\,576$ cells, i.e. about one million.
5. $\dfrac{N}{N_0} = 10^6$, so $g = \dfrac{\log(10^6)}{\log(2)} = \dfrac{6}{\log(2)} \approx 19.93$: 20 complete generations are needed. Time: $20 \times 30 = 600$ minutes, i.e. about 10 hours.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 225 and the rate or common ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 22 — **Review and assessment**

Common logarithm function — 10 Dec 2026 • B: 1 h – C: 2 h

Objectives.

- Summarise the methods.
- Identify the conditions.
- Correct mistakes.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for an increase of 25 %. $1 + \dfrac{25}{100} = 1.25$
2. Expand $\log(3b)$, $b > 0$. $\log 3 + \log b$
3. Give the domain of log. $]0, +\infty[$
4. Calculate $\log(10^4)$. 4
5. Solve $10^x = 0.01$. $x = -2$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: summarise the methods and identify the conditions.

Activity 8. Make a map of methods: domain; properties; equations $10^x = b$; equations $a^x = b$; inequalities. Attach a condition and an example to each branch.

Solution. The map should in particular show that arguments must be positive and the distinction between a base greater than 1 and one between 0 and 1. Have students say aloud which piece of data triggers the choice of method.

Written record. Before calculating a logarithm: check positivity. Before a comparison: identify the direction of variation. Before a conclusion: check that the result is plausible.

Example 9. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. The map should in particular show that arguments must be positive and the distinction between a base greater than 1 and one between 0 and 1.

Exercise 3.8 · essential.

1. Calculate $\log(10^4)$ and $\log(10^{-4})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.

3. Solve $4^x = 24$ and give an approximate value to the nearest hundredth.

Solution. 4 and -4. The expression equals $2 + 2\log(a) - \log(b)$. Finally $x = \dfrac{\log(24)}{\log(4)}$.

Further practice: Calao, pp. 76–79. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. We look for a method to find the number of digits of a natural number $N \geqslant 1$ without writing it out in full. The integer part of a real number $y \geqslant 0$ is the greatest integer less than or equal to y; it is denoted $E(y)$.

1. Let $N = 4\,718$.
   a. Bound N between two consecutive powers of 10. Deduce bounds for $\log(N)$ between two consecutive integers.
   b. Compare $E(\log N)$ with the number of digits of N.
2. An integer N is written with 15 digits.
   a. Bound N between two consecutive powers of 10, then $\log(N)$ between two consecutive integers.
   b. What is $E(\log N)$?
3. State a rule giving the number of digits of N from $\log(N)$. Test it on $N = 1\,000$.
4. Use this rule to find the number of digits of: a. 3^{100}; b. 2^{64}; c. 365×10^{15}.
5. A password is made of 12 characters, each chosen from 62 symbols (26 lower-case letters, 26 upper-case letters, 10 digits). There are 62^{12} possible passwords. How many digits does this number have?

Solution. **Target:** Number of digits of an integer.

1. a. $10^3 \leqslant 4\,718 < 10^4$. Since log is strictly increasing, $3 \leqslant \log(4\,718) < 4$ (we find $\log(4\,718) \approx 3.674$).
   b. $E(\log N) = 3$ and N has 4 digits: number of digits $= E(\log N) + 1$.
2. a. The smallest 15-digit integer is 10^{14} and the smallest 16-digit integer is 10^{15}: $10^{14} \leqslant N < 10^{15}$, so $14 \leqslant \log(N) < 15$.
   b. $E(\log N) = 14$, and $14 + 1 = 15$ digits.
3. If N has k digits, $10^{k-1} \leqslant N < 10^k$, so $k - 1 \leqslant \log(N) < k$: $E(\log N) = k - 1$. Rule: *the number of digits of N is $E(\log N) + 1$*. For $N = 1\,000$: $\log(1\,000) = 3$, $E(3) + 1 = 4$ digits, which is correct.
4. a. $\log(3^{100}) = 100\log(3) \approx 47.71$: $47 + 1 = 48$ digits.
   b. $\log(2^{64}) = 64\log(2) \approx 19.27$: 20 digits (indeed $2^{64} = 18\,446\,744\,073\,709\,551\,616$).
   c. $\log(365 \times 10^{15}) = \log(365) + 15 \approx 17.56$: 18 digits.
5. $\log(62^{12}) = 12\log(62) \approx 21.51$: 62^{12} is written with 22 digits (about 3.2×10^{21} passwords).

Homework. Redo practice exercise 3 with the following data: replace the initial value by 230 and the rate or common ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.